

Foundation Programme
Universiti Teknologi Malaysia
Session 2020/2021 Semester 3

Test 2 (20%)

Time: 90 minutes

FSPM 0024 Calculus

Answer all SEVEN questions. Show all your work clearly. Any similarity with answers given by apps/chegg will be punished accordingly. List of formula is given on the next page.

1. Given $\int_1^6 f(x) dx = -9$ and $\int_1^6 g(x) dx = \begin{cases} x^2 - 2, & 1 \leq x \leq 4 \\ 3x + 2, & 4 \leq x \leq 6 \end{cases}$.

(a) Evaluate $\int_1^6 [2f(x) - g(x)] dx$.

(3 marks)

(b) Find the value of k that satisfies $\int_6^1 [2g(x) - kf(x)] dx = -1$.

(3 marks)

2. By using substitution $u^2 = 3x^2 + 1$, evaluate $\int_0^1 \frac{4x^3}{\sqrt{3x^2 + 1}} dx$.

(5 marks)

3. (a) Use integration by parts to find $\int x \sin(3x + 1) dx$.

(3 marks)

(b) Using answer in part (a), find $\int x^2 \cos 3x dx$.

(3 marks)

4. By using substitution $u = \sin x$, evaluate $\int \frac{\cos x}{2 \sin^2 x + 3 \sin x + 1} dx$ using partial fractions.

(6 marks)

5. By taking $x = 2 \cos \theta$, show that

$$\int_0^1 \sqrt{4 - x^2} dx = -4 \int_{\frac{\pi}{2}}^{\frac{\pi}{3}} \sin^2 \theta d\theta.$$

Hence, evaluate $\int_0^1 \sqrt{4 - x^2} dx$.

(7 marks)

$$1) \int_1^6 f(x) dx = -9 \quad \int_1^6 g(x) dx = \begin{cases} x^2-2 & 1 \leq x \leq 4 \\ 3x+2 & 4 \leq x \leq 6 \end{cases}$$

$$\begin{aligned} (a) \quad 2 \int_1^6 f(x) - \int_1^6 g(x) &= 2(-9) - \left(\int_4^6 3x+2 + \int_1^4 x^2-2 \right) \\ &= -18 - \left[\frac{3x^2}{2} + 2x \right]_4^6 + \left[\frac{x^3}{3} - 2x \right]_1^4 \\ &= -18 - (66 - 32) + \left(\frac{4^3}{3} - \frac{2}{3} \right) \\ &= -18 - (34 + 15) \\ &= -67 \end{aligned}$$

$$(b) \int_6^1 2g(x) - k f(x) dx = -1$$

$$-2 \int_1^6 g(x) dx + k \int_1^6 f(x) dx = -1$$

$$-2(49) + k(-9) = -1$$

$$-9k = -1 + 98$$

$$k = -\frac{97}{9}$$

$$2) \quad u^2 = 3x^2 + 1, \quad \frac{d}{dx}(u^2) = \frac{d}{dx}(3x^2 + 1), \quad 2u \frac{du}{dx} = 6x, \quad dx = \frac{2u du}{6x} = \frac{2u}{6x} du$$

$$\int_0^1 \frac{4x^3}{\sqrt{3x^2+1}} dx = 4 \int_0^1 \frac{x^3}{\sqrt{u^2}} \cdot \frac{2u}{6x} du = \frac{4}{3} \int_0^1 x^2 du = \frac{4}{3} \int_0^1 \left(\frac{u^2-1}{3} \right)$$

$$= \frac{4}{9} \int_0^1 u^2 - 1 du = \frac{4}{9} \left[\frac{u^3}{3} - u \right]_0^1 = \frac{4}{9} \left[\frac{(\sqrt{3x^2+1})^3}{3} - \sqrt{3x^2+1} \right]_0^1$$

$$= \frac{4}{9} \left(\frac{2}{3} + \frac{2}{3} \right) = \frac{4}{9} \left(\frac{4}{3} \right) = \frac{16}{27}$$

$$3) (a) \int x \sin(3x+1) dx, \quad u = x, \quad \frac{dv}{dx} = \sin(3x+1)$$

$$= uv - \int v \frac{du}{dx} dx \quad \frac{du}{dx} = 1, \quad v = \frac{-\cos(3x+1)}{3}$$

$$= x \left(\frac{-\cos(3x+1)}{3} \right) - \int \frac{-\cos(3x+1)}{3} dx$$

$$= \frac{-x \cos(3x+1)}{3} + \frac{\sin(3x+1)}{9} + C$$

$$= \frac{\sin(3x+1) - 3x \cos(3x+1)}{9} + C$$

$$(b) \int x^2 \cos 3x dx = uv - \int v u' = x^2 \left(\frac{\sin 3x}{3} \right) - \int \frac{\sin 3x}{3} (2x) = \frac{x^2 \sin 3x}{3} - \frac{2}{3} \int x \sin 3x$$

$$u = x^2 \quad v' = \cos 3x$$

$$= \frac{x^2 \sin 3x}{3} - \frac{2}{3} \left(\frac{-x \cos 3x}{3} - \int \frac{-\cos 3x}{3} \right)$$

$$u' = 2x \quad v = \frac{\sin 3x}{3}$$

$$= \frac{x^2 \sin 3x}{3} + \frac{2x \cos 3x}{9} - \frac{2 \sin 3x}{27} + C$$

$$4) \quad u = \sin x, \quad \int \frac{\cos x}{2\sin^2 x + 3\sin x + 1} dx = \int \frac{\cancel{\cos x}}{2u^2 + 3u + 1} \cdot \frac{du}{\cancel{\cos x}} = \int \frac{1}{2u^2 + 3u + 1} du = \int 2\left(\frac{1}{2u+1}\right) - \frac{1}{u+1} du$$

$$\frac{du}{dx} = \cos x$$

$$dx = \frac{du}{\cos x}$$

$$\frac{1}{2u^2 + 3u + 1} = \frac{1}{(2u+1)(u+1)} = \frac{A}{2u+1} + \frac{B}{u+1} = \frac{2}{2u+1} - \frac{1}{u+1}$$

$$A(2u+1) + B(u+1) = 1$$

$$(A+2B)u + A+B = 1$$

$$A+2B=0, \quad A+B=1$$

$$A=-2B, \quad -B=1$$

$$A=2, \quad B=-1$$

$$= 2 \int \frac{1}{2u+1} - \int \frac{1}{u+1}$$

$$= \ln 2u+1 - \ln u+1$$

$$5) \quad x = 2\cos \theta \quad \int_0^1 \sqrt{4-x^2} dx = \int_0^1 \sqrt{4-(2\cos \theta)^2} (-2\sin \theta) d\theta = \int_0^1 \sqrt{4-4\cos^2 \theta} (-2\sin \theta) d\theta$$

$$\frac{dx}{d\theta} = -2\sin \theta$$

$$dx = -2\sin \theta d\theta$$

$$\text{when } x=1, \quad \theta = \cos^{-1}\left(\frac{1}{2}\right) = 60^\circ = \frac{\pi}{3}$$

$$\text{when } x=0 \quad \theta = \cos^{-1}(0) = 90^\circ = \frac{\pi}{2}$$

$$= \int_0^1 2\sin \theta (-2\sin \theta) d\theta$$

$$= -4 \int_0^1 \sin^2 \theta d\theta$$

$$= -4 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sin^2 \theta d\theta$$

$$= -4 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{1}{2} (1 - \cos 2\theta) d\theta$$

$$= -2 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 1 - \cos 2\theta d\theta$$

$$= -2 \left[\theta - \frac{\sin 2\theta}{2} \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

$$= -2(0.614 - 1.571)$$

$$= 1.914$$

$$6) \quad x = y^2 - 4y + 2, \quad x = y - 2$$

$$x = y^2 - 4y + \left(\frac{4}{2}\right)^2 - \left(\frac{4}{2}\right)^2 \quad x=0, y=2 \quad (6)$$

$$x = (y+2)^2 - 2 \quad y=0, x=-2 \quad (7)$$

$$\text{min: } (-2, 2) \quad (1)$$

$$y-2 = y^2 - 4y + 2$$

$$y^2 - 5y + 4 = 0$$

$$(y_1 = 4, y_2 = 1) \quad (3)$$

$$(x_1 = 2, x_2 = -1)$$

$$\text{(intersection point)}$$

$$\text{when } x=0, y^2 - 4y + 2 = 0$$

$$y_1 = 0.586, y_2 = 3.414 \quad (4) \quad (5)$$

$$\text{Area} = \int_1^4 y^2 - 4y + 2 - y + 2$$

$$= \int_1^4 y^2 - 5y + 4$$

$$= \left[\frac{y^3}{3} - \frac{5y^2}{2} + 4y \right]_1^4$$

$$= \left(\frac{4^3}{3} - \frac{5(4^2)}{2} + 16 \right) - \left(\frac{1}{3} - \frac{5}{2} + 4 \right)$$

$$= -\frac{5}{3} - \frac{1}{6} = -\frac{11}{6}$$

$$7) y = x^2, y = 2x + 8$$

$$x^2 = 2x + 8$$

$$x^2 - 2x - 8 = 0$$

$$x = 4, x = -2$$

$$\text{Volume} = \pi \int_{-2}^0 (2x + 8)^2 - x^4 dx$$

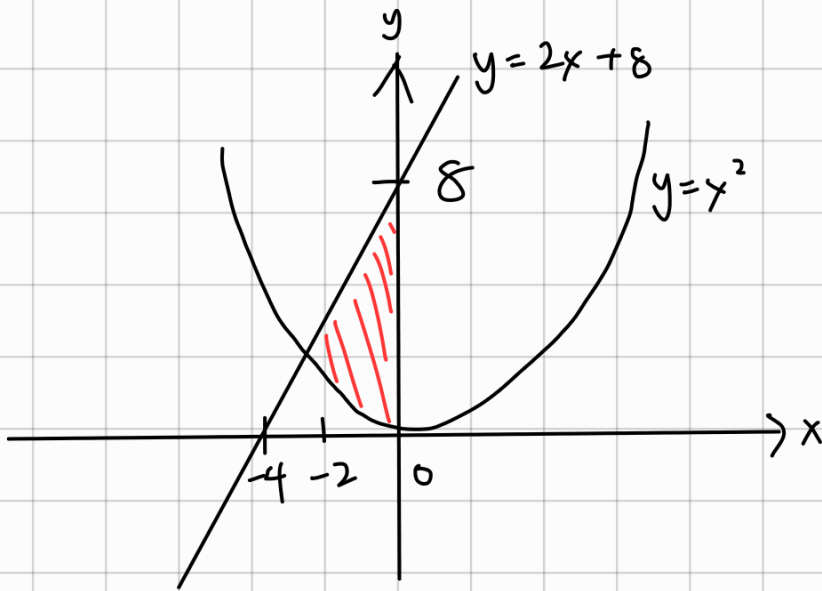
$$= \pi \int_{-2}^0 4x^2 - 32x + 64 - x^4 dx$$

$$= \pi \int_{-2}^0 -x^4 + 4x^2 - 32x + 64 dx$$

$$= \pi \left[-\frac{x^5}{5} + \frac{4x^3}{3} - 16x^2 + 64x \right]_{-2}^0$$

$$= \pi (0 - (-196.27))$$

$$= 196.27 \pi \text{ unit}^3$$



6. Sketch and find the area of a region bounded by the curve $x = y^2 - 4y + 2$ and the line $x = y - 2$.

(5 marks)

7. Sketch the region that is enclosed by the curve $y = x^2$ and the line $y = 2x + 8$. Find the volume of solid revolution formed by revolving the region 360° about the x -axis.

(5 marks)

LIST OF FORMULA

Trigonometric	
• $\cos^2 x + \sin^2 x = 1$ • $1 + \tan^2 x = \sec^2 x$ • $\cot^2 x + 1 = \operatorname{cosec}^2 x$ • $\sin 2x = 2 \sin x \cos x$ • $\cos 2x = \cos^2 x - \sin^2 x$ $= 2 \cos^2 x - 1$ $= 1 - 2 \sin^2 x$ • $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$	• $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$ • $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$ • $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$ • $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$ • $2 \sin x \sin y = -\cos(x + y) + \cos(x - y)$ • $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$
Differentiations	Integrations
$\frac{d}{dx}[k] = 0, k \text{ constant}$	$\int k dx = kx + C, k \text{ constant}$
$\frac{d}{dx}[x^n] = nx^{n-1}$	$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$
$\frac{d}{dx}[e^x] = e^x$	$\int e^x dx = e^x + C$
$\frac{d}{dx}[\ln x] = \frac{1}{x}$	$\int \frac{dx}{x} = \ln x + C$
$\frac{d}{dx}[\cos x] = -\sin x$	$\int \sin x dx = -\cos x + C$
$\frac{d}{dx}[\sin x] = \cos x$	$\int \cos x dx = \sin x + C$
$\frac{d}{dx}[\tan x] = \sec^2 x$	$\int \sec^2 x dx = \tan x + C$
$\frac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x$	$\int \operatorname{cosec}^2 x dx = -\cot x + C$
$\frac{d}{dx}[\sec x] = \sec x \tan x$	$\int \sec x \tan x dx = \sec x + C$
$\frac{d}{dx}[\operatorname{cosec} x] = -\operatorname{cosec} x \cot x$	$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$